

Comments about the notes:

For this course, you are allowed to share the questions that you have submitted for the tests. Therefore, there was made in secret a doc that was shared with all students taking the course, where you should share your questions and then the correct answer for them.

This helped A LOT, and I would recommend you guys do it again. The way we did it was going on Learn → My Course → Class list → Select all → Deselect all the TA's and Teachers and then sent an email with the link.

We also posted the link in the GE group chat. Anyways, these are all the questions that students submitted on the docs.

Third Multiple Choice Test

Introduction:

The purpose of this document is to collect and share questions and answers with everyone for the multiple-choice test 😊. Please follow the format below when adding your questions. If your question is very similar to an existing one, please come up with a different one, since the professor will likely change duplicates during the test. All multiple-choice questions will be stored here. Please **save this link** so you can return and add future questions.

Please make sure your questions cover the entire module, as failing to do so will increase the number of questions the professors have to submit themselves.

Quote from learn:

“In this exam we encourage calculation questions, comparative analysis, or story/problem based questions. “

Format:

Group #: (Group members

Question: xxx

A Answer A

B Answer B

C Answer C (the correct answer)

D Answer D

... (add three questions)

Third Multiple Choice Test

Group 61:

Question: What primarily causes the sharp increase in effort required as recycling rates approach very high levels?

- A Rising consumer demand for secondary materials
- B Higher energy efficiency of recycling equipment
- C Reduced availability of primary raw materials
- D **The remaining material becomes increasingly hard to collect, separate, and purify**

Question: In a material system where both end-of-life scrap and pre-consumer scrap enter recycling, what aspect most strongly determines how much secondary material actually substitutes virgin material?

- A The functional quality of the recycled material relative to the virgin material**
- b The lifetime of stock held in buildings and infrastructure
- c The number of recycling facilities in a region
- D The level of waste exports to other countries

Question: In recycling system analysis, why can high values of stock accumulation reduce the share of secondary materials entering production, even if recycling efficiency is good?

- A Stock accumulation increases overall collection rates
- B Longer retention of materials in use delays their return as recyclable flows
- C Stock accumulation boosts the economic value of recycled goods**
- D Stock accumulation reduces contamination in waste streams

Group 30:

In which country is the majority of rare earth minerals extracted?

- a) China**
- b) India
- c) USA
- d) Venezuela

Which of the following is a major benefit of improving resource recycling?

- a) It increases the amount of raw materials that need to be mined
- b) It reduces energy use and environmental impact**
- c) It makes products harder to manufacture
- d) It decreases the lifespan of materials

What is the purpose of the TRL scale?

- A. To measure the environmental impact of industrial processes

B. To assess the level of technological readiness of a technology

C. To estimate the economic cost of implementing new materials

D. To classify waste streams based on recycling potential

Group 37

Question: Which of the following does not describe closed-loop recycling?

A A product is recycled to be used in a different product

B There is little to no need for new material input

C It is also referred to as a "feedback system"

D The material purity is maintained and known

Question: Which is not one of the main purposes of resource recycling?

A To reduce dependence on foreign raw materials and strengthen geopolitical stability

B To support economic efficiency and resource conservation

C To increase the consumption of new natural resources

D To minimize environmental impacts such as pollution and waste generation

Question: Which of the following best describes the concept of the *Old Scrap Ratio (OSR)* in recycling systems?

A The proportion of recycled material that originates from production waste

B The share of recycled input that comes from end-of-life (post-consumer) products

C The amount of material that cannot be recycled and becomes landfill waste

D The efficiency rate of recycling processes in industrial production

E The percentage of total material demand covered by newly extracted resources

Group 7

Question: The *Old Scrap Ratio* is critically impacted by which of the following factors?

A Increases in recycling efficiency resulting in less new scrap

B Improved waste sorting technology.

C Change of metal prices in the short term which reduces profitability.

D Illegal waste trade which causes misallocation of flows, distorted reporting and bypass of regulated recycling systems.

Question: Which of the following is least critical to recycling in circularity?

A Physico-chemical composition of materials.

B Use of new products, end-of-life management of materials

C Consumer awareness campaigns and packaging labels

D Substitution of alternative production of consumption of resources.

Question: Which of the following isn't a part of the "second law of thermodynamics" perspective on recycling?

- A Closing the loop
- B Circular economy
- C Energy extraction via incineration**
- D Cradle-to-cradle

Group #50 -  BOMBOCLAAT 

Question: What is a key characteristic of "downcycling"?

- a) The reprocessed material has an increased value and functionality.
- b) The process avoids the production of a virgin material.
- c) The recycled material experiences a loss of quality and functionality.**
- d) It is exclusively part of a closed-loop recycling system.

Question: Which major factor influences recycling systems in context of "China Waste Import Ban"?

- a) The crucial influence of market demands and international trade.**
- b) The critical role of technology in material separation.
- c) The importance of product design for disassembly.
- d) The environmental benefits of reducing transportation emissions.

Question: According to the case study, why is phosphorus recycling difficult in Denmark?

- a) The recycling machines aren't good enough.
- b) Farmers can use other fertilizers instead.
- c) The problem is getting the phosphorus to the farms that need it.**
- d) Denmark doesn't have enough phosphorus in its waste.

Group #52

Question: In an Open Material Loop, the purity of a product:

- A Always remains pure
- B Can vary depending on the source materials and processes**
- B Increases because additional material is added
- D Always decreases over time

Question: By the year 2030, the European Commission aims to achieve a packaging waste recycling of:

- A 55%
- B 60%
- C 75%

D 70%

Question: What does ORER stand for?

- A Overall Resource Extraction Rate
- B Optimal Recycling Extraction Ratio
- C Overall Recycling Efficiency Rate**
- D Optimal Reuse and Recovery

Group 5

Question: what does the Old Scrap Ratio (OSR) measure?

- A The ratio between recycled content and total materials produced
- B The share of recycled metals in global trade flows
- C The fraction of recycled end-of-life (post-consumer) material in total recycling flows**
- D The percentage of manufactured products designed for recyclability

Question: what was Denmark's actual municipal solid waste (MSW) recycling rate in 2023?

- A 66%
- B 25%
- C 47%**
- D 58%

Question: what characterizes downcycling?

- A The recycled material loses quality and functionality compared to the original** It increases the amount of raw materials that need to be mined
- B The recycled material gains higher value and improved properties
- C The material is reused with identical quality and functionality
- D The material becomes suitable for closed-loop recycling only

Group #23

Question: Which stage in the circularity system most directly influences the quality and functional performance of recycled materials that enter new production cycles?

- A Physico-chemical composition of materials, such as alloying elements or additives that determine recyclability
- B Manufacturing of new products, including resource consumption, emissions, and formation of residual fractions
- C Source separation, collection, and transport of discarded materials
- D Production of secondary raw materials through refinement and processing steps that determine material properties**

Question: According to the data on European plastic-waste exports from 2016 to 2018, which country shows a sharp decline in received plastic waste around early 2018, followed by increased exports to several Southeast Asian countries?

- A India
- B Vietnam
- C China**
- D Turkey

Question: Which of the following is identified as a general problem that can undermine the reliability of recycling indicators?

- A Overestimation of resource efficiency
- B Excessive focus on pre-consumer waste
- C Illegal trading of materials**
- D Lack of economic domains in indicator design

Group 4 (Aidan, Erick, Advait, Ines)

Question: Why is the RC (recycled content) so high for Pb?

- A There are strict international bans on mining, forcing industries to rely only on recycled lead
- B There are low extraction rates and large amounts in the network**
- C The metal becomes more efficient when reused multiple times, improving its performance with each cycle
- D The strong consumer demand for “green” lead products pushes to recycle far more

Question: What are the 3 major Waste Management Perspectives?

- A Planning, Manufacturing, Decaying
- B Recycling, Reusing, Incinerating
- C Overall processing, Production, End-of-life**
- D Manufacturing, Selection, Product Withdrawal

Question: Why is TRL a relevant tool?

- A It provides a method of evaluating possible expansion locations of a technology.
- B It allows the consumer to provide feedback on an upcoming technology.
- C It provides a framework to assess the readiness of a technology for release into the market**
- D It assesses the manufacturing capacity to create a technology.

Group 3

Question:

The difference between upcycling, recycling and downcycling is best described by what statement?

- A. Upcycling and recycling reduce the quality of materials, whereas downcycling increases their value.
- B. Recycling always produces a product identical to the original, whereas upcycling and downcycling both result in low-quality materials.
- C. Value or functionality is increased through upcycling, recycling keeps similar quality, and downcycling results in reduced quality and functionality.**
- D. Recycling always requires chemical processes, while upcycling and downcycling involve only mechanical processes.
- E. Upcycling applies only to plastics, recycling only to metals, and downcycling only to organic waste.

Question:

A recycling facility processes 600 tonnes of post-consumer end-of-life (EoL) scrap and 400 tonnes of pre-consumer scrap.

Using the OSR formula:

$$OSR = \frac{r}{r + k}$$

What is the Old Scrap Ratio (OSR)?

- A. 0.30
- B. 0.40
- C. 0.60**
- D. 0.75
- E. 0.10

Question:

Denmark's waste composition is distributed as follows:

- 40% from construction & demolition (C&D)
- 29% from households
- 14% from the service sector
- 8% from industry
- 9% from power plants and wastewater sources

Which sector provides the greatest opportunity for increasing national recycling rates, and why?

- A. Households, because they generate the most contaminated waste streams
- B. Construction & demolition, because they account for the largest share of total material waste**
- C. Industry, because industrial waste is always hazardous

- D. Power plants, because of the waste being already pre-sorted
- E. The service sector, because it produces mainly recyclable high-value materials that are easy to recover

Group 15

Q1. Which shape represents the model that defines a “safe and just operating space” between a social foundation and an environmental ceiling?

- A) **Doughnut**
- B) Pyramid
- C) Spiral
- D) Ladder

Q2. How many levels are on the scale for TRLs?

- A) **9**
- B) 5
- C) 7
- D) 11

Q3: How many levels are on the scale for MRLs?

- A) **12**
- B) 8
- C) 10
- D) 14

Group 20

Question: Which statement is a valid reason that hinders society from reaching 100% recycling rate?

- A **The cost of effort is greater than the benefits seen from recycling more than about 75%.**
- B According to the predator-prey theory, a recycling rate above 75% would increase the extraction of primary resources, which would decrease the recycling rate.
- C The benefits of it would be too high, which would lower motivation of future world leaders to innovate new technologies.
- D It is impossible because we will never have the necessary infrastructure

Question: What does the End-of-life processing rate known as EoL PR show?

- A **How efficient is the process of taking one material and turning it into another**
- B how efficient is the process of recycling processed material back into raw material

- C how much material we recycle and not dump in the landfills
- D how much material we process into less environmentally harmful before dumping them in the landfills

Question: Which of the following is highly affected by trading?

- A End-of-life processing rate
- B **End-of-life recycling rate**
- C Overall recycling efficient rate
- D Overall processing rate

Group 25

Question 1 — Based on Danish waste statistics, approximately what percentage of annual waste in Denmark comes from construction and demolition (C&D) activities?

- A. 14%
- B. 29%
- C. **40%**
- D. 75%

Question 2 — According to the lecture, why is recycled plastic often more contaminated than virgin plastic?

- A. Plastics naturally degrade during their use phase.
- B. Recycling facilities mix plastics with metals during processing.
- C. **Plastic waste contains both physical and chemical contaminants.**
- D. Additives evaporate during the washing stage.

Question: 3. According to the updated Waste Framework Directive, what is counted as entering a “final recycling process”?

- a) **Waste that requires no further mechanical sorting and enters a production process to be reprocessed**
- b) Any waste that is collected for potential recycling
- c) Waste that has been sorted but still requires separation into materials
- d) Waste that is incinerated with energy recovery

Group 13

Question: What happens when humanity crosses a planetary boundary?

- A) Nothing significant changes
- B) The planet immediately becomes uninhabitable
- C) The risk of large-scale environmental changes increases**
- D) Economic growth stops

Question: What does the Doughnut model aim to show?

- A) How to increase global GDP
- B) How to meet human needs within planetary limits**
- C) How countries can compete economically
- D) How to reduce taxes worldwide

Question: What has happened to global ecological overshoot since 2000?

- A) It has decreased
- B) It has stayed the same
- C) It has increased**
- D) It has disappeared

Group 17

Question: What is NOT the purpose of recycling indicators?

- A) Provide a quantitative measure
- B) Concisely describe the STATE of a system
- C) Communicate information about a system
- D) Directly control or change the system**

Question: What is the new calculation method for recycling targets?

- A) i/e**
- B) g/e
- C) $i/(i+k)$
- D) $(i+k)/(g+k)$

Question: How much are global CO₂ eq emissions expected to increase if electricity were brought to the 19% of the world who lives without it?

- A 18%
- B 25%
- C 1%**
- D 5%

Group 27

Question: Which of the following is a primary consequence of urbanization related to the production and consumption of goods?

- A Increased reliance on renewable energy sources
- B Large material flows and large quantities of waste in cities**
- C Reduced demand for agricultural products
- D Decreased air pollution levels

Question: Which of the following is a primary consequence of industrialization related to product composition?

- A Introduction of more complex products containing many chemicals and compounds**
- B Decrease in the use of raw materials
- C Reduction in global trade of manufactured goods
- D Increased reliance on hand-made products

Question: According to the Waste Framework Directive (2008/98/EC), how is waste defined?

- A Waste is any substance or object that has been abandoned by the holder
- B Waste refers only to non-recyclable materials
- C Waste is any substance or object which the holder discards or intends or is required to discard**
- D Waste includes only hazardous substances

Group 60

Question: Which of the following resources does China produce the vast majority of?

- A Petroleum
- B Uranium
- C Rare Earth Metals**
- D Copper

Question: What is a major problem with the recycling of plastics?

- A It is impossible
- B It requires inputs of rare materials during the recycling processes
- C It mostly produces low quality plastics which there is little demand for in the global market**
- D It causes release of massive amounts of microplastics during the recycling processes

Question: What is the formula for Recycling Input Rate (RIR)?

- A $(\text{secondary input})/(\text{virgin input})$
- B $(\text{recycled output} + \text{upcycled output})/(\text{total input})$
- C $(\text{secondary input})/(\text{total input})$**
- D $(\text{recycled input})/(\text{recycled input} + \text{upcycled input} + \text{downcycled input})$

Group 42

Question: Which of these product designs shows the highest single product recycled material value and also a low degree of material mixing and is therefore very lucrative in recycling? (Gutowski et. al., 2008)

- A) Aluminium can
- B) Catalytic converter**
- C) Automobile battery
- D) Laptop
- E) Glass bottle

Question: Calculate the Recycling Input Rate (RIR) as the fraction of secondary material in the total input with the following quantities for metals: EoL metal $i=50$ kt ; Pre-consumer waste $j=30$ kt ; Primary metal input $a=100$ kt ; metal content in EoL products $e=110$ kt ; EoL metal collected $g=70$ kt

- A) 57% (ORER)
- B) 64% (CR e/g)
- C) 44% (RIR)**
- D) 45% (RR i/e)
- E) 71% (PR i/e)

Question: Which are the only two planetary environmental boundaries that, according to new 2025 data (Potsdam Institute for Climate Impact Research (PIK)), were not exceeded in 2025?

- A) Ocean acidification & Freshwater Depletion
- B) Landsystem change & Introduction of Novel Entities
- C) Air pollution & Stratospheric Ozone Depletion**
- D) Change in Biosphere Integrity & Climate Change

Group 11

Question: According to the data for Denmark, which sector generates the largest proportion of waste?

- A) Households
- B) Industry
- C) Service Sector
- D) Construction & Demolition**

Question: What is identified as a major reason for the current waste problems we face today?

- A) Lack of Technology
- B) Urbanization and Industrialization**

- C) Decreased Consumer Awareness
- D) Strict Government Regulations

Question: According to the waste statistics presented for Denmark (Disposal 2020), what was the approximate percentage of the total waste that was sent to landfilling?

- A 75%
- B 25%
- C 3%**
- D 9%

Group # 9

Question: What are the main factors that reduce the quality of recycled plastic and make closing the loop difficult?

- A High mechanical strength of plastics
- B Physical and chemical contamination in the waste stream**
- C Excessively long product lifetime
- D Too much demand for recycled plastic

Question: How is the Old Scrap Ratio defined?

- A Share of virgin material in production
- B Fraction of pre-consumer waste recycled
- C Fraction of recycled end-of-life material in total recycled flows**
- D Ratio of secondary metal prices to primary metal prices

Question: What is the key difference between closed-loop and open-loop recycling?

- A Closed-loop always requires heat, open-loop does not
- B Closed-loop creates products of higher value than the original
- C Closed-loop returns materials to the same product system, open-loop sends them to different product systems**
- D Open-loop recycling can only be used for metals

Group 18

Question: In a metal recycling system, the Old Scrap Ratio (OSR) is defined as

OSR = (end-of-life scrap) / (end-of-life scrap + new scrap).

Which action would directly increase the OSR of the system?

- A: Increasing the amount of primary (virgin) metal used while keeping scrap flows constant.
- B: Replacing parts of the end-of-life scrap with a additional new (pre-consumer) scrap.

c: Sending more end-of-life scrap to landfill instead of recycling.

d: Collecting more end-of-life scrap while keeping the amount of new scrap the same.

Question: A smelter receives 30 kt of end-of-life scrap (i), 20 kt of pre-consumer scrap (j), and 50 kt of primary metal (a) in one year.

Using the definition

$$\text{RIR} = (i + j) / (i + j + a)$$

What is the Recycling Input Rate (RIR)?

A: 20%

B: 40%

C: 60%

D: 50%

Question: A high-quality plastic used for food-grade bottles is recycled into low-grade plastic used for park benches with lower performance requirements. Which term best describes this change in material quality?

A: Upcycling.

B: Downcycling.

C: Closed-loop recycling.

D: Waste prevention.

Group 55:

1: Overall processing rate (OPR) is defined as

a) the share of post-consumer waste which is collected compared to the potential collectable.

b) the fraction of post-consumer that is collected, pre-treated, recycle.

c) the efficiency of waste separation/preparation prior to production/manufacturing.

d) the efficiency of post-consumer waste processing.

2: According to the Doughnut Economics model, what lies inside the Safe and Just Operating Space?

- A) The Environmental Ceiling
- B) The Social Foundation
- C) Critical Human Deprivation**
- D) Climate Change Threshold
- E) Global Income Distribution

3: According to Denmark's 2023 municipal solid waste data, what was the actual recycling rate for total MSW?

- A. 58%
- B. 47%**
- C. 37%
- D. 75%

Group 16

Question: Which region does the EU mainly depend on for phosphorus imports?

- A) Central America and the Caribbean
- B) Central Asia and Eastern Europe
- C) West and Central Africa
- D) Middle East and North Africa**

Question: Which of the following is not an example of physical contamination in plastic waste?

- A) Non-plastic materials, missortings
- B) Non-plastic materials, non-separable
- C) Non-targeted plastic
- D) Substances introduced through manufacturing**

Question: Which of the following are all environmental boundaries as defined by Doughnut Economics?

- A) Biodiversity breakdown, Climate change, and Nutrient pollution**
- B) Social cohesion, Land conversion, and Peace and justice
- C) Political voice, Connectivity, and Ocean acidification
- D) Income and work, Chemical pollution, and Water and food

Group 21

Question: A recycling plant processes a metal that has a very long product lifetime (30–50 years). What is the primary consequence for the recycling system?

- A) The Old Scrap Ratio becomes artificially high

B) The recycling system receives end-of-life scrap only long after demand for secondary material has already increased

C) The Recycling Input Rate automatically increases

D) The material can only be recycled through open-loop routes

Question: A company redesigns a product to reduce the number of material types from 8 to 3 while keeping the same functionality. What is the most likely effect on recycling performance?

A) The OSR decreases because fewer materials enter the system

B) Recycling efficiency increases due to reduced material mixing

C) The EoL processing rate drops because complexity is reduced

D) The Recycling Input Rate becomes independent of primary material demand
RIR

Question: A city introduces a new waste-sorting system that increases the collection of end-of-life scrap from households by 25%, without changing industrial scrap flows. What is the expected effect on recycling indicators?

A) The Old Scrap Ratio increases

B) The New Scrap Ratio increases

C) The Overall Recycling Efficiency Rate decreases

D) The Recycling Input Rate becomes zero

Group 8:

Question: Which of the following sectors produce the most waste?

A Service sector

B Industry

C Households

D power plants and wastewater treatment

Question: What is NOT a purpose of indicators?

A Provide a quantitative measure

B Concisely describe the state of a system

C Describe the processes of a system

D communicate information about a system

Question: What is part of the production perspective

A Overall processing rate (OPR)

B Overall recycling efficiency rate (ORER)

C Recycling input rate (RIR)

D End-of-life recycling input rate (EoL RIR)

Group 33:

Question: Which sector percentually recycles the most of its waste in Denmark (2019-2023)?

- A Construction & demolition
- B Households
- C Industry**
- D Service

Question: Which sector percentually incinerates the most of its waste in Denmark (2019-2023)?

- A Construction & demolition
- B Industry
- C Households**
- D Service

Question: What best describes the Recycling Input Rate (RIR)?

- A The fraction of secondary input compared to virgin material in a product
- B The fraction of potential secondary input in a product
- C The numerical or percentual fraction of secondary input compared to the total material input in a product**
- D Qualitatively decided by independent panel

Group 29:

In the regional material cycle, which additional element must be accounted for compared to the global material cycle?

- A. Natural dissipation to the environment
- B. Import and export of natural resources, products, and secondary materials**
- C. Lifetime assumptions of metal products
- D. Stocking and destocking effects

Which of the following best reflects the overarching purpose of recycling indicators as described in the lecture?

- A. To determine the exact economic value of recovered materials
- B. To describe the state of a material system concisely and facilitate communication**
- C. To enforce regulatory compliance directly on recycling companies
- D. To eliminate the need for tracking stock changes in material flow analysis

Which of the following pairs contains one item from the "Social Foundation" and one from the "Environmental Boundaries"?

- A. Peace and Justice & Social Cohesion
- B. Food & Nutrient Pollution**

- C. Education & Political Voice
- D. Housing & Land Conversion

Group 10 -

Question: When comparing a traditional linear economy model ("take-make-dispose") to a circular economy model within management modeling, which of the following is a fundamental distinction in how they approach resources and waste?

- A The linear model prioritizes cost reduction, while the circular model prioritizes job creation.
- B The linear model relies on government regulation, while the circular model is driven solely by consumer choice.
- C The linear model treats waste as an externality to be managed, while the circular model designs waste out of the system by viewing it as a resource.**
- D The linear model is only applicable to manufacturing, while the circular model applies only to digital services.

In a circular material system, which factor most directly *reduces* the Recycling Input Rate (RIR) even when recycling technologies are efficient?

- A. An increase in post-consumer scrap collection
- B. A decrease in product complexity leading to easier disassembly
- C. High market demand for virgin materials driving up primary material input**
- D. Longer product lifetimes reducing annual scrap flows

Question: The city of "EcoVille" is tracking its progress toward circular economy goals. In a given year, the city generated 100,000 metric tons of municipal solid waste. Through various programs, 35,000 metric tons were collected for recycling, and 15,000 metric tons were collected for composting.

Based on this data, what is EcoVille's diversion rate?

- A 15%
- B 35%
- C 50%**
- D 65%

Group 12-

Question: Which recycling indicator is least affected by the economic market mechanisms (like price fluctuations or product lifespan) and is therefore considered the most relevant measure of a society's technical performance in waste management?

- A. Old Scrap Ratio
- B. Recycling Input Rate
- C. Recycled Content

D. End-of-Life Recycling Rate

Question: Based on Danish waste statistics which show that the largest fraction of waste comes from the Construction and Demolition (C&D) sector (40%), followed by Households (29%), why is the C&D sector considered the greatest opportunity for increasing national recycling rates?

A. C&D waste is typically already sorted by material type at the source, leading to minimal contamination and high recycling efficiency.

B. C&D waste accounts for the largest mass share of total waste, meaning improvements here have the highest impact on overall national rates.

C. The C&D sector uses mainly non-toxic, easily separable materials, which simplifies the end-of-life processing rate (EoL PR).

D. Household waste incineration rates are too low to allow for significant material recovery from that stream.

Question: The Substance Concentrating Efficiency (SCE_j) indicator measures how a system transforms the concentration and mass flow of a substance j.

If a system results in an SCE value that is negative $SCE_j < 0$, what is the practical outcome of the process on substance j?

A. The process achieves its most optimal recycling state.

B. The substance is diluted or dispersed within the system.

C. The substance has been concentrated into a purer output stream.

D. The concentration of the substance remains perfectly unchanged.

Groupe 14:

Question: A company collects 1000 kg of end-of-life (EOL) plastic products from consumers (e). After sorting and separation, 800 kg are selected for recycling (g). Finally, 600 kg of recycled plastic actually enter the market (i). What is the End-of-Life (EOL) Recycling Rate?

A. 0.8

B. 0.6

C. 0.75

D. 1.0

Question: A company producing smartphones is struggling with rising material costs and increasing waste from discarded devices. Most phones are thrown away after only

two to three years, even though many components are still functional. To strengthen circularity and improve resource cycling, which strategy should the company prioritize?

- A. Increase advertising for the newest models
- B. Use single-use adhesives to seal components permanently
- C. Introducing modular designs allowing parts to be replaced**
- D. Improving packaging to reduce shipping-related material use

Question: What are the current major challenges we face today regarding waste management?

- A. Lack of urbanization and low industrial activity
- B. Global decrease in population density
- C. Worldwide elimination of plastic production
- D. Urbanization and industrialization**

Group 35:

Question: How do you calculate an EoL processing rate?

- a) EoL recycled/EoL products
- b) EoL collected/EoL products
- c) EoL recycled/EoL collected**
- d) EoL products/EoL collected

Question: What are the first three top action of the optimized waste hierarchy look?

- a) prevention, high quality recycle, recycling with reversable loss of properties
- b) high quality recycling, recycling with reversable loss of properties, recycling to a non circular product
- c) prevention, high quality recycle, landfill
- d) prevention, preparation for reuse, high quality recycle**

Question: Which of the nine planetary boundaries have we not yet crossed?

- a) atmospheric aerosol loading, stratospheric ozone depletion**
- b) freshwater change, stratospheric ozone depletion

- c) Ocean Acidification, freshwater change
- d) atmospheric aerosol loading, ocean acidification

Group 40: (Lucia, Matilde, Hugo, Chris)

What is the End-of-Life Recycling Rate?

- a) The fraction of a material collected for recycling from the waste stream compared to its total availability in end-of-life waste.
- b) The percentage of recycled material used in the manufacturing of a new product.
- c) The mass of recycled material that actually becomes new products, divided by the mass of material collected for recycling.
- d) The total amount of scrap processed by a recycling facility in a given year.

What is the recycling input rate?

- A the fraction of secondary material in the total input
- B The fraction of post-consumer waste collected compared to the potential collectable
- C The fraction of recycled end-of-life scrap in the recycling flows
- D The fraction of post-consumer material in the total input

What is the old scrap ratio, OSR, formula:

- A $i / (i+k)$
- B $k + i / (k-i)$
- C $i+k / (i)$
- D $i/(i-k)$

Second Multiple Choice Test

Introduction:

The purpose of this document is to collect and share questions and answers with everyone for the multiple-choice test 😊. Please follow the format below when adding your questions. If your question is very similar to an existing one, please come up with a different one, since the professor will likely change duplicates during the test. All multiple-choice questions will be stored here. Please **save this link** so you can return and add future questions.

Please make sure your questions cover the entire module, as failing to do so will increase the number of questions the professors have to submit themselves.

Format:

Group #: (Group members)

Question: xxx

A Answer A

B Answer B

C Answer C (the correct answer)

D Answer D

... (add three questions)

Second Multiple Choice Test

Group #50: (Elhenas, Kharlos, Matheos, Mhartinos) - [Greek Mafia](#)

Question: What is one of the primary purposes of conducting a historical analysis of resources? flows

A To prove that economic development is impossible without increasing resource consumption.

B To predict the exact date when specific resources will be completely depleted.

C To identify past trends and their drivers in order to inform future scenarios and interventions.

D To assign blame to specific countries or industries for historical environmental damage.

Question: What is the main characteristic of dynamic Material Flow Analysis (MFA) compared to traditional MFA?

A It only looks at resource flows at a single point in time

B It focuses on the chemical composition of materials

C It considers how resource flows and stocks change over time

D It ignores emissions and outputs from human-made systems

Question: The analysis shows that as global resource consumption has increased, what has happened to the circularity of the economy?

A Circularity rates have decreased significantly, leading to a linear economy.

B Circularity rates have increased at the same rapid pace as consumption.

C Circularity rates have remained relatively consistent, despite an increase in circularity potential.

D Circularity has become the dominant model, successfully decoupling growth from resource use.

Group 37:

Question: What are the factors in the IPAT-equation?

A) I is impact, P is population, A is affluence and T is technology

B) I is impact, P is precipitation, A is affluence and T is technology

C) I is impact, P is population, A is APAC and T is teflon

D) I is impurities, P is production, A is avoided waste and T is technology

Question: How is the environmental learning rate calculated?

$eLR(\%) = (1 - 2^{-\alpha_e}) * 100\%$. Where α is the power constant.

$eLR(\%) = (1 - 2^{T_d}) * 100\%$. Where T_d is the technology doubling time

$T_d(a) = t * \frac{\ln 2}{\ln(1 + \frac{r}{100})}$. Where r is the growth rate and t is time.

$T_d(a) = t * \frac{\ln 2}{\ln(1 + \frac{r}{100})}$. Where r is the discount rate and t is technology.

fun

Question: What is EPR

- A) Extended Production Recycling
- B) **Extended Producer Responsibility**
- C) Energy production report
- D) Energy production rate

Group 42:

Question: What is the polymer that generates the highest primary global plastic waste (62 million tonnes

annually)?

- A) **Polypropelene**
- B) Polyethylenterephthalat
- C) Low-Density Polyethylen
- D) High-Density Polyethylen
- E) Polyvinylchloride

Question: What is meant by Ecological Cycling?

- A) **Ecological cycling implies that outputs of biomass re-enter biogeochemical cycles and contribute to new plant growth.**
- B) The anthropogenic impact and alteration of the natural nitrogen and carbon cycles
- C) Circular biomass flows from ecosystems contributing to new biomass generation in the human-made technosphere
- D) The cycling of resources in a circular economy, to be able to substitute virgin production and close material circularity

Question: What could be examples for a functional recycling product from construction wood?

- A) **Fiberboard or chipboard for construction**
- B) Wood shaving for use as animal and pet bedding
- C) Wood fiber thermal insulation panels
- D) Wood pellets for incineration in residential heating

**Group
27:**

Question: What is the main function of the Cas9 protein in the CRISPR/Cas gene-editing system?

- A) It recognizes and binds to mRNA molecules to regulate translation.
- B) It cuts DNA at a specific location determined by a guide RNA sequence.**
- C) It inserts new DNA sequences randomly throughout the genome.
- D) It replicates the DNA strands to create additional gene copies.

Question : In a Retrospective Top-Down Dynamic MFA (used in the SS Earth model), what is the main characteristic of this approach?

- A) Uses future trends to predict possible scenarios.
- B) Calculates inflows and outflows using lifetime distributions.
- C) Analyzes past stock changes based on existing input-output data.**
- D) Focuses on forecasting material demands for 2050 and beyond.

Question : According to Pivnenko (2016), which mitigation measure is considered the most effective long-term strategy for reducing chemical contaminants in material cycles?

- A) Constrain (selective recycling)
- B) Remove (improved decontamination)
- C) Prevent (phase-out or substitution)**
- D) Destroy (incineration)

Group 25: (Mikkel - Valentina - Rasmus - Hákun)

Question: What is the primary advantage of Prospective LCA compared to Conventional LCA?

- A) It ignores future uncertainties
- B) It evaluates environmental impacts of future technologies and systems**
- C) It focuses solely on economic impacts
- D) It uses less data and fewer assumptions

Question: What is the main distinction between primary and secondary microplastics?

- A) Primary are biodegradable, while secondary are not
- B) Primary are formed during polymerization, secondary during photosynthesis
- C) Primary are manufactured as small particles, while secondary result from degradation of larger plastics**
- D) Primary occur naturally, secondary are synthetic

Question: What is downcycling in material recycling?

- A) Reusing materials without changing their quality
- B) Recycling that completely closes material loops
- C) Recycling that produces lower-quality materials, only partially closing material loops**
- D) Recycling materials to create higher-grade products

Group 7:

1. **Question: The process of recycling can lead to lower material quality leading to only partial closure of the material cycle. How is this “downcycling” effect mitigated currently?**

- a) **integrated use of virgin material to increase material quality to specification.**
- b) chemical enhancement during recycling process to raise quality.
- c) lowest quality recyclable waste is taken to landfill to maintain higher recycling quality.
- d) micro-organisms are used to break down lowest quality into other compounds.

2. **Question: Which is the correct order of the qualitative hierarchy order?**

- a) **Prevent, Constrain, Remove, Destroy, Contain**
- b) Prevent, Contain, Remove, Destroy, Constrain
- c) Remove, Prevent, Constrain, Destroy, Contain
- d) Contain, Constrain, Prevent, Destroy, Remove

3. **Question: Which sort of chemicals/agents would you expect from surface protectant additives:**

- a) plasticizers, lubricants, blowing agents
- b) **antistatic agents, antifriction agents, adhesion improving agents, anti-fog.**
- c) flame retardants, fillers/reinforcement materials
- d) pigments or colourants

Group 5:

Question: Which of the following best describes a **chemical impurity** in recycling system

- A Foreign materials such as caps or labels attached to bottles
- B Plastic fragments sorted into the wrong material stream
- C Substances added intentionally to improve production, like pigments and stabilizers**
- D Small paper pieces mixed into glass waste

Question: Why does **downcycling** (recycling into lower quality materials) only partially close material loops?

- A Because it completely eliminates impurities
- B Because high-quality materials no longer need virgin inputs
- C Because low-quality recyclates still depend on virgin materials to maintain function**
- D Because it always increases substitution efficiency

Question: Which measure is the **most effective long-term way** to reduce harmful chemicals in material cycles?

- A Contain – isolate materials in landfills
- B Remove – improve decontamination processes

- C Prevent – phase out or ban the chemical
- D Constrain – recycle only selected fractions

Group 20:

Question: How does Dynamic MFA differ from MFA:

- A. Dynamic MFA introduces location into system
- B. There is no significant difference
- C. Dynamic MFA introduces time concept into system
- D. Dynamic MFA can introduce figures from the past

Question: What does it mean when a chemical is pre-registered?

- A. The chemical was previously on the market but got pulled and is being reviewed again
- B. The chemical passed some of the safety tests but not all yet
- C. The chemical's safety and environmental impact has been tested before entering the market.
- D. The chemical passed some of the safety tests and promised to pass the rest in near future

Question: What are NIAS?

- A. Non-Important Added Substances
- B. Non-Intentionally Accumulated Substances
- C. Non-Intentionally Added Substances
- D. Non-Integrated Additional Substances

Group 60: (Antonia, An-Li , Igor , Olgierd)

Question: Which statement best describes why downcycling only partially closes material loops?

- A) Downcycled materials cannot be recycled again
- B) High-quality applications still require virgin materials when recycling produces lower quality outputs
- C) Downcycling always results in energy recovery
- D) Downcycled materials have higher impurity levels than virgin materials

Question: What is the primary distinction between primary and secondary microplastics?

- A) Primary microplastics are biodegradable, while secondary are not.
- B) Primary microplastics are intentionally manufactured at a small size, while secondary microplastics result from the breakdown of larger plastic items.
- C) Primary microplastics are found in the ocean, while secondary microplastics are found on land.
- D) Primary microplastics are larger than 5mm, while secondary microplastics are smaller than 5mm.
- E) Primary microplastics are those that are recycled, while secondary microplastics are destined for landfill.

Question: Which of the following is NOT a motivation to carry out historical analysis?

- A) Show the impacts of global events on resource consumption trends
- B) Aid in developing future scenarios
- C) Understand the trends of the past
- D) Quantify the current distribution of materials within a product's life cycle**

Group 13: (Elin, Helene, Ida, Rikke)

Question: Which of the following statements is NOT a reason for conducting historical analysis in resource engineering?

- A) Understand trends of the past
- B) Show the impacts of global events on trends
- C) To predict future events with complete accuracy**
- D) Help us develop future scenarios

Question: Why is it relevant to consider chemicals in goods?

- A) Most goods today are produced without any chemical substances
- B) It's not
- C) Chemicals are present in all products and goods**
- D) Chemical impurities are automatically removed during production

Question: Which process interrupts the circular flow of materials and quality circularity?

- A) Production
- B) Waste Management
- C) Landfill or incineration**
- D) Distribution

Group 30:

Question: In the context of circularity, why are chemical impurities in recycled materials a major challenge?

- A. They improve the mechanical strength of the material.
- B. They make the recycling process cheaper and faster.
- C. They have no impact on the recyclability of materials.
- D. They degrade material quality and limit the number of recycling loops possible.**

Question: What is a key environmental concern discussed in the lecture on microplastics?

- A. Their contribution to improving soil fertility.
- B. Their persistence in aquatic environments and potential bioaccumulation in food chains.**
- C. Their ability to neutralize chemical pollutants in water.
- D. Their short degradation time under sunlight exposure.

Question: What stage often has the highest environmental impact for most products?

- A. Distribution and retail phase.
- B. Disposal phase only.
- C. Raw material extraction and production phase.**
- D. Customer use phase has no measurable impact.

Group 14:

Question: Regarding the hierarchy of mitigation measures for chemicals, why should the options “Destroy” and “Contain” be avoided when possible?

- A) Because they completely eliminate the source of emissions.
- B) Because they are difficult and expensive to implement.
- C) Because they increase waste production**
- D) Because they prevent the generation of hazardous substances at the source.

Question: What is the maximum size that defines microplastics (MPs) ?

- A) < 10 mm
- B) < 5 mm**
- C) < 1 mm
- D) < 0.5 mm

Question: What is the main limitation of recycling materials to lower qualities (downcycling)?

- A) It requires additional purification steps to reach the same quality as virgin materials.
- B) It can only lead to a partial closing of material cycles.**
- C) It increases production costs.
- D) It allows the material to be reused indefinitely without significant quality loss.

Group 4: (Ines, Aidan, Advait, Erick)

Question: Define what a Prospective life cycle assessment(pLCA) is?

- A) A forward looking assessment that focuses on the environmental impacts of products that are already on the market.
- B) A simplified version of LCA that only considers energy consumption by future life forms during production.
- C) A forward looking extension of LCA that evaluates environmental impacts of emerging technologies, products, or systems.**
- D) An analysis of past environmental performance for existing industrial processes

Question: What is a pre-registered chemical?

- A) One which has been patented before entering the market.
- B) One which is using technology from the past.
- C) One which is released into the market with a low initial cost.
- D) One which has done all the required environmental and safety testing before launching into the market.**

Question: What is typically the maximum tolerance of impurities accepted in a recycling center?

- A) 2%**
- B) 5%
- C) 8%

D) 12%

Group 52: (Marius, Mathias, Tobias, Mads)

Question: What is a pLCA?

- A) An extension of the LCA, which solely look at products and their impact throughout it's entire lifetime.
- B) An LCA model that evaluates the impact of by-products derived from the extraction- and use of the main product.
- C) A forward looking extension of LCA that evaluates environmental impacts of emerging technologies, products, or systems.**
- D) A backwards looking extension of LCA that strictly evaluates impact of processes a technology, product or system has gone through.

Question: What happens to material reuse when downcycling occurs?

- A) The material loop is completely closed, so no new raw materials are needed.
- B) It gives the biggest environmental benefit because it always saves the most material.
- C) It only partly keeps materials in use, since high-quality products still need new raw materials.**
- D) It means the material gets dirty only from chemicals added during processing.

Question: How is Microplastics defined?

- A) Plastics measuring <10 mm in their biggest dimension
- B) Plastics measuring <1 mm in their biggest dimension
- C) Plastics measuring <5 mm in their biggest dimension**
- D) Plastics measuring <0,1 mm in their biggest dimension

Group 9: (Andrea, Hakan, Silas)

Question: which of the following is NOT a problem when measuring microplastics

- A) lack of standardized methods
- B) size limits for the methods used to measure
- C) identification challenges
- D) finding a sample containing microplastics.**

Question: what technology is used to identify plastic polymers?

- A) X-ray
- B) IR Spectroscopy (FTIR/LDIR). NIR**
- C) Electron Microscopy
- D) DNA Sequencing

Question: Which source produces large microfiber emissions?

- A) Car exhausts
- B) Industrial laundry.**

- C) Food packaging
- D) Waste incinerators

Group 3 (Asta, Amalie, Belma & Younis):

Question: A metal can is found in the paper waste. What kind of impurity is this?

- A) **Missorting**
- B) Interfering material
- C) Lower grade material
- D) Chemical Impurity

Question: Plastic film has printed colours and logos that need to be removed. What kind of impurity is this?

- A) Missorting
- B) Interfering material
- C) Lower grade material
- D) Chemical Impurity**

Question: A PET bottle has a small metal handle attached. What kind of impurity is this?

- A) Missorting
- B) Interfering material**
- C) Lower grade material
- D) Chemical Impurity

Group 16

Question: In a Dynamic Material Flow Analysis (Dynamic MFA), what distinguishes a retrospective top-down approach from other dynamic methods?

- A) It predicts future resource use based on assumed behavioral changes.
- B) It calculates inflows and outflows using lifetime distributions of materials.
- C) It analyzes past resource stocks and flows using existing aggregate input-output data.**
- D) It models material flows at a product level using bottom-up lifetime estimations.

Question: Why does downcycling only lead to a partial closing of material loops in the context of circularity?

- A) Because downcycled materials completely replace virgin materials in all applications.
- B) Because downcycled materials retain identical functionality and purity as virgin materials.
- C) Because high-quality applications still require virgin materials due to loss of material quality and functionality**
- D) Because impurities are entirely removed during recycling, ensuring total circularity.

Question: Which of the following statements best explains why prevention (chemical phase-out or bans) is considered the most effective long-term measure for reducing chemical impurities in circular material systems?

- A) Because prevention eliminates the need for any recycling infrastructure or monitoring of chemical flows.
- B) **Because prevention directly removes hazardous chemicals from production cycles, stopping their accumulation through multiple recycling loops.**
- C) Because prevention immediately cleans existing waste streams of all legacy contaminants.
- D) Because prevention relies solely on advanced decontamination technologies to treat existing waste stocks.

Group 11

Q: Question: What is the main environmental benefit of recycling materials?

- A) It increases the number of waste management facilities
- B) **It substitutes the production of virgin materials**
- C) It eliminates all material losses in the recycling process
- D) It ensures all recycled products are of higher quality

Question: Which factor most directly determines how well a material loop can be closed in a circular economy?

- A) The number of recycling facilities available
- B) **The functionality and quality of the recycled material compared to the virgin material**
- C) The total amount of waste collected per year
- D) The type of packaging used during product distribution

Question: What is the main purpose of a prospective life cycle assessment?

- A) To evaluate the current environmental performance of existing products
- B) **To forecast environmental impacts of future technologies and systems**
- C) To compare the costs of different production methods
- D) To measure historical data on resource use and emissions

Group 13

Question: Which world region is responsible for the biggest contribution to GHG per capita?

- A) West Asia
- B) **North America**
- C) Europe
- D) Eastern Europe, Caucasus and Central Asia
- E) Asia and the Pacific

Question: Which term explains what happens to a product when it is recycled into a lower quality product?

- A) **Downcycling**
- B) Functional recycling
- C) Material cascading
- D) Secondary recycling

E) Upcycling

Question: Which of the following is not a so-called impurity in the material loop?

- A) Missortings
- B) Problematic impurities**
- C) Interfering materials
- D) Lower grade materials
- E) Chemical impurities

Group 29

Q1 - What is the difference between a primary and secondary microplastics?

- A) Primary microplastics are intentionally added to products and are more homogeneous, while secondary microplastics are decomposed debris from a bigger plastic**
- B) Primary microplastics are smaller than 5mm and larger than 1mm, while all microplastics smaller than that are secondary microplastics
- C) We call primary microplastics those debris that suppose a larger environmental threat than the secondary ones.
- D) Primary microplastics have a long degradation life (hundreds of thousands of years), while secondary plastics have a shorter one (thousands of years)

Q2 - Which of the following statements about impurities is false?

- A) Chemical contaminants can be introduced during the use-phase
- B) Lower grade material is not always recyclable
- C) Interfering materials impurities are easily separable with initial manual control**
- D) Chemical impurities can be unintentionally added through the production process

Q3 - Which statement best reflects global resource trends since 1970?

- A) Population has doubled, GDP has decreased, and resource use has remained constant
- B) Population has doubled, GDP has increased significantly, and resource extraction has risen roughly 3.5x**
- C) Population has remained constant while circularity has doubled
- D) Resource use has decreased as technology improved

Group 10:

Question: What are the four types of impurities that occur when recycling?

- A) Organic, Inorganic, Hazardous, and Electronic waste
- B) Cross-contamination, Mechanical Hazards, Quality Reducers, and Toxic Leachates.
- C) Missortings, Interfering Materials, Lower Grade Materials, and Chemical Impurities.**

D) Pre-consumer, Post-consumer, Industrial, and Agricultural residuals.

Question: What's the main reason why downcycling does not fully close material loops in a circular economy?

- A) Because it always increases the amount of recycled material available
- B) Because downcycled materials have higher energy content than virgin materials
- C) Because the recycled material loses quality and cannot substitute high-quality virgin material
- D) Because downcycling completely removes all impurities from the materials

Question: In the context of material circularity, what does “functionality” refer to?

- A) The amount of energy required to recycle a material
- B) The visual appearance of the recycled material
- C) The physical and chemical properties that allow a material to perform its intended purpose
- D) The market value of recycled material compared to virgin material

Group 17

What is an important factor in how much material cycles can be closed?

- A) The quality of material
- B) The amount of material
- C) The complexity of the material
- D) The country of origin of the material

What is the relationship between social well-being and income levels when examining resource footprints across regions?

- A) Higher-income regions tend to have lower material footprints because of efficiency and technology
- B) Higher-income regions generally show higher resource consumption per person, while lower-income regions rely less on resource-intensive sectors
- C) Social well-being increases as resource use decreases in all regions
- D) Resource footprints are evenly distributed across all income levels and sectors
- E) There is no clear connection between resource use and social well-being across different regions

Why is black plastic a problem for sorting machines?

- A) Sorting machines use infrared light, which can recognize translucent plastics but not black ones.

- B) Black plastic is heavier than other plastics, so it damages the sorting equipment.
- C) The color black absorbs too much moisture, confusing the sorting sensors.
- D) Sorting machines use xray, which can recognize translucent plastics but not black ones.
- E. Black plastic contains magnetic pigments that interfere with the sorting system.

Group 8 (Samuel, Jannik, Sturla, Alfred)

Which alternative best describes downcycling?

- A) **When a recycled material has lower quality than the virgin material**
- B) When a recycled material is used in another manufacturing process than what it was extracted for
- C) When a product can straight away be used again without any need for processing
- D) When a recycled material has higher quality than the virgin material

Which two regions have the highest material productivity?

- A) **North America & Europe**
- B) Europe & West Asia
- C) North America & West Asia
- D) West Asia & Africa

What is the correct ordering in the qualitative hierarchy of mitigation measures?

- A) **Prevent > Constrain > Remove > Destroy > Contain**
- B) Destroy > Remove > Prevent > Constrain > Contain
- C) Constrain > Prevent > Destroy > Contain > Remove
- D) Destroy > Contain > Remove > Constrain > Prevent

Group 40: (Lucia, Matilde, Hugo, Chris)

Question: What is a dynamic MFA?

- A) Introduces location into the model
- B) Introduces energy consumption associated with flows
- C) Introduces time into the model**
- D) Introduces material flow in the past, but not present or future.

Question: In which area is plastic most commonly used?

- A) **Packaging**

- B) Building and construction
- C) Transportation
- D) Textile sector

Question: Which of the following is a cause of impurities:

- A) **Missorting**
- B) Substitution
- C) Functional recycling
- D) Functionality

Group 35:

Question: In which regions of the world is the sectoral material footprint of food over 1/3 of all material consumption

- A) Africa, APAC, EECCA
- B) Africa, APAC, LAC
- C) Africa, EECCA, LAC**
- D) West Asia, APAC, LAC
- E) Africa, West Asia, EECCA

Question: What statement about the impurity „lower grade material“ is wrong?

- A) it is the targeted material, but with unwanted properties
- B) Can technically be recycled
- C) Needs extra processing step for Identifying their presence and a treatment phase for their removal**
- D) May cause loss of quality of the recycled material or decrease sorting efficiency

Question: Which of the following statements about plastics is **not** correct?

- A) They are light and can provide energy savings, for example in transport.
- B) They are versatile and can be shaped and molded, with properties modified by chemical substances.
- C) They are waterproof and hygienic, making them useful in many applications.
- D) They have good insulation properties and are biodegradable, helping reduce environmental impact.**

Group 21

In the context of chemical impurities and circularity, what is the most effective long-term strategy for reducing hazardous chemicals in material cycles?

- A) Remove – through improved decontamination processes
- B) Constrain – by recycling only selected fractions
- C) Prevent – by phasing out or substituting problematic chemicals
- D) Contain – by isolating materials in landfills

In a Prospective Life Cycle Assessment (pLCA), what is the primary purpose of incorporating dynamic background data and socioeconomic pathways?

- A) To account for how technologies and systems may evolve in the future
- B) To eliminate uncertainty from LCA modeling
- C) To focus exclusively on static datasets from current systems
- D) To measure only direct energy consumption in manufacturing

According to the Dynamic Material Flow Analysis (Dynamic MFA) lecture, what distinguishes this method from a static MFA?

- A) It introduces time, allowing analysis of how stocks and flows change over a period
- B) It excludes emissions and focuses only on resource inflows
- C) It models materials without considering historical data
- D) It only considers spatial (geographical) variations of material use

Group 18

In a dynamic material flow analysis (MFA), what does “dynamic” aspect primarily refer to?

- A) Adding geographic detail to the system model

- B) Tracking how stocks and flows change over time
- C) Excluding emissions to focus only on material inputs
- D) Mapping regional differences at a single point in time (purely spatial modeling)

Why do prospective LCAs often use scenario pathways (e.g., SSPs)?

- A) To reflect plausible future changes in technologies and background systems
- B) To remove all uncertainty from the model
- C) To restrict the study to present-day datasets only
- D) To replace sensitivity analysis entirely by fixing one “official” future

Which example best fits ecological cycling?

- A) Compost returning nutrients to soil and supporting new biomass growth
- B) Regrinding PET flakes to substitute virgin PET in bottles
- C) Burning wood pellets for energy recovery
- D) Landfilling organic waste to isolate it from the environment

Group 23

What are missortings (impurities) in material sorting processes?

- A) Materials that have been fully refined and purified
- B) Material fractions that do not match the targeted sorting guidelines
- C) Materials that increase process performance and recycling yield
- D) Materials that are intentionally added to improve recycling efficiency

According to the qualitative hierarchy of mitigation measures, which action focuses primarily on preventing contaminants?

- A) Phasing out substances with or without substitution
- B) Using waste incineration to destroy contaminants
- C) Applying new decontamination technologies to remove contaminants
- D) Disposing of waste in landfills to contain contaminants

What best describes functional recycling of substances?

- A) Elemental properties are utilised in a new material
- B) Elements are lost during processing and cannot be recovered
- C) Utilisation of properties is negligible or non-existent
- D) Materials are down-cycled and dispersed in construction materials

Group 55

Which of the following strategies represents the most effective long-term approach to reducing the accumulation of hazardous chemicals in material recycling systems?

- A. Increasing the energy input to improve decontamination efficiency
- B. Restricting recycling to selected high-quality material fractions
- C. Banning or phasing out problematic chemicals at the production stage *
- D. Storing contaminated materials in controlled landfills

Which of the following statements about recycling quality and circularity is correct?

- A. All recycling processes close the material loop completely, regardless of quality loss.
- B. Downcycling improves material functionality and fully substitutes virgin materials.
- C. The environmental benefit of recycling comes from substituting virgin materials.*
- D. Impurities in waste materials usually increase the recyclability of materials.
- E. Chemical impurities are always easily removed during recycling.

Which of the following is NOT an impurity in recycling?

- A. Mis-sortings
- B. Lower grade material
- C. Interfering materials
- D. Recyclable base material

Group 1

Which region of the world consistently recorded the highest per capita greenhouse gas emissions in 1970, 2000, and 2020?

- A. North-America
- B. EECCA (Eastern Europe, Caucasus, and Central Asia)
- C. APAC (Asia-Pacific)
- D. LAC (Latin America and the Caribbean)
- E. Europe

Which type of impurity needs special treatment in order to be sorted?

- A. Missortings
- B. Interfering materials
- C. Chemical contaminants

D. Lower grade materials

What is the main source of primary microplastic in the environment?

A. Abrasion of tyres through driving

B. Laundering of synthetic clothes

C. Intentionally added microplastics in personal care products

D. Road markings

Group 6 (Somaya, Mariam, Jeton & Khalil)

Which of the following best describes secondary microplastics?

A. Plastic particles intentionally produced for use in cosmetics

B. Microplastics that result from the breakdown of larger plastic items

C. Microplastics that are used as abrasives in industrial cleaning

D. Nanoplastics smaller than 1 μm

What is the most effective long term measure to reduce hazardous chemicals in material cycles?

A. Improved recycling technology

B. Restricting use to certain waste fractions

C. Chemical substitution with similar compounds

D. Preventing or banning the chemical

What does a substitution ratio below 1 (e.g., 0.8) indicate?

A. Recycled material performs better than virgin

B. Recycled material performs equally to virgin

"C. Recycled material has lower functionality and quality than virgin"

D. Recycling leads to full circularity

Group 1:

What processes has to be done to turn plastic waste into recycled plastic?

A. Shredding, caustic washing, density separation, drying, extrusion & pelletization

B. Decontamination, extrusion & pelletization, enrichment, remolding

C. Sorting, Shredding, caustic washing, reactivation, carburization

D. Petrification, extrusion, caustic washing, purification, quality selection

What does **not** describe doughnut economy?

- A. Doughnut economy seeks to avoid the current systems reliance on constant economic growth
- B. Doughnut economy aims to be between a social foundation and ecological ceiling
- C. Doughnut economy uses social needs and environmental impacts as parameters
- D. Doughnut economy incentivises economic growth through increased production

When can the output of any sorting operation be counted as recycled according to the new EU waste directive from 2008?

- A. when the waste streams for disposal or incineration remain below 10%
- B. when the waste streams for disposal or incineration remain below 20%
- C. when the waste streams for disposal or incineration remain below 5%
- D. when the waste streams for disposal or incineration remain below 30%

First Multiple Choice Test

Introduction:

The purpose of this document is to collect and share questions and answers with everyone for the multiple-choice test 😊. Please follow the format below when adding your questions. If your question is very similar to an existing one, please come up with a different one, since the professor will likely change duplicates during the test. All multiple-choice questions will be stored here. Please **save this link** so you can return and add future questions.

Please make sure your questions cover the entire module, as failing to do so will increase the number of questions the professors have to submit themselves.

Format:

Group #: (Group members)

Question: xxx

A Answer A

B Answer B

C Answer C (the correct answer)

D Answer D

... (add three questions)

First Multiple Choice Test

Group 37:

Question: What characterizes the information and digital age compared to the previous?

a. Plastics are categorized for themselves

b. Introduction of Mechanized Coal and Steam Powered Production Facility

c. Enabled by rail expansion, transportation, telephones, clean water, and electricity

d. Upended economic markets leading to global financial crises throughout the 1920s and 1930s

Question: Which of the following is not a source of biomass

a. Livestock manure

b. Sewage sludge

c. Energy crops

d. Wind

e. wood

Question: What is a good in an MFA?

- a. economic entities of matter with either a positive or negative value. Often consist of one or more of the most basic flow type in an MFA
- b. any (chemical) element or compound composed of uniform units
- c. Amount of materials stored in a process, this can be temporary or long term. transformation, transport, or storage of flows in an MFA

Group 23:

Question: What is the characteristic of a "fund resource"?

- A A resource that can be used, but not depleted.
- B A resource that exist as finite, fixed amounts in the natural environment, with renewal rates beyond human timescales.
- C A resource that can be depleted at a rate depending on the ratio between extraction and renewal.
- D A resource which cannot be found in nature, it only exists through human production.

Question: How is flux in an MFA measured ?

- A In output units per input units
- B In input units per output units
- C In units per time
- D It is dimensionless

Question: What is NOT a correct challenge of using "secondary raw materials" ?

- A Their composition is not always known
- B Not always accessible
- C It cannot be implemented in a circular system
- D Has varying quality standards

Group 14:

Question: In a copper mining operation, silver is also recovered during the extraction process. Silver in this case is considered a:

- A Primary product
- B Co-product
- C By-product
- D None of the above, because in copper mining only copper can be extracted

Question: Which of the following are defined as bioresources?

- A Fossil fuels
- B Abiotic resources
- C Forests
- D All of the above

Question: Which energy resource has the highest share in Denmark?

- A Coal
- B Solar
- C Gas
- D Wind**

Group 3:

Question: You map many plausible futures for aluminum demand without picking the most likely one.

- A Predictive
- B Normative
- C Explorative**
- D Ex-post

Question: You estimate Denmark's 2035 e-waste based on trends and lifetimes. What scenario analysis are you using?

- A Normative
- B Predictive**
- C Explorative
- D Ex-post

Question: You start from a 95 % waste-reduction target and work backward. What scenario analysis are you using?

- A Normative**
- B Predictive
- C Explorative
- D c

Group 4:

Question: Which of the following is considered a model of weak sustainability?

- A Sustainable development goals
- B Principles of sustainability
- C Triple bottom line**
- D Absolute sustainability

Question: Which of the flow would be a flow in an MFA

- A 30 liters of gasoline in a fuel tank
- B Boiling rice
- C Raw copper going to a factory**
- D Construction of laptops

Question: What is the goal of industrial ecology

- A investigate the use of resources to reduce the environmental impacts of their use

B Study the stocks and flows of resources to make more sustainable and equitable manufacturing processes

C Study the stocks and flows of resources to make more sustainable and equitable systems that come closer to closing material loops

D Investigate the use of resources to such that they are deniably consumed by closing material loops

Group 5:

Question: What does the transfer coefficient (TC) in an MFA model describe?

A The amount of time a material stays inside a stock

B The economic value of a good

C The percentage of a substance transferred from one input into a specific output good

D The spatial size of a system boundary

Question: In Material Flow Analysis (MFA), the term goods refers to:

A Any (chemical) element or compound composed of uniform units

B Transformation, transport, or storage of flows

C Economic entities of matter with positive or negative value

D The amount of materials stored in a process

Question: What is the main goal of increasing circularity in material flow analysis

A To speed up fossil fuel extraction

B To increase the energy content of natural gas

C To reduce the use of virgin raw materials and minimize waste

D To measure only the short-term stocks of materials

Group 6:

Which of the following is a correct example of a *renewable resource*?

A) Fossil fuels (coal, oil, gas)

B) Minerals and ores

C) Solar radiation

D) Uranium (nuclear fuel)

E) Copper

In Material Flow Analysis (MFA), what is a *substance*?

A) Economic entities of matter with a positive or negative value (e.g. drinking water, fuel oil, waste)

B) Any (chemical) element or compound composed of uniform units, homogeneous in constitution (e.g. N, C, Cu, CO₂)

C) An umbrella term that includes both substances and goods, such as raw or modified materials

- D) A transformation, transport, or storage of flows in an MFA system
- E) The amount of materials stored in a process, either temporary or long-term

Why are bioresources not automatically carbon neutral, despite being renewable?

- A) Because bioresources contain heavy metals that increase emissions.
- B) Because their cultivation and use compete with land, food, and energy demands, and indirect land use change can cause additional CO₂ release.
- C) Because they are chemically similar to fossil fuels.
- D) Because their carbon cycle requires thousands of years to renew.

Group 7:

Question: Which of the following could be considered bioresources:

- A) Resources derived from living organism that renew in short biological timescales.
- B) Abiotic resources
- C) Fossil fuels (coal, oil, gas etc. formed from ancient life)
- D) Water, soil, minerals etc.
- E) Resources from living organisms.

Question: What is the primary goal of industrial ecology?

- A) Study the stocks and flows of goods for economic optimisation.
- B) Study the stocks and flows of goods to design more efficient manufacturing systems.
- C) Study the stocks and flows of resources to design more sustainable and equitable systems which close material loops.
- D) Study the stocks and flows of resources to design more sustainable and equitable systems that come closer to closing the materials loop.

Question: Which of the following isn't an Abiotic resource?

- A) Iron
- B) Ground water
- C) A plant
- D) Wind

Group 8:

Question: What do we address mainly in ex-post studies?

- A) model-induced uncertainty
- B) Model, Measurement, & Parameterization Uncertainty
- C) accounting/comparing
- D) data-uncertainties

Question: Which country is the largest supplier of critical raw materials?

- A) China

- B) Brazil
- C) USA
- D) India

Question: When is a commodity considered a “by-product”?

- A) When the commodity has plenty of economic value and is worth extracting by itself.
- B) When the commodity is recovered in connection to extracting a primary commodity and provides a comparatively small economic value.
- C) When the commodity has zero economic value.
- D) When the supply of the commodity is always elastic.

Group 9:

Question: Which of the statements is not about strong sustainability?

- A It states that there are no essential differences between natural and manmade capital
- B It states that natural capital can't be fully substituted by manufactured capital
- C It focuses on inter- and intragenerational needs rather than increasing stock and capital
- D It focuses on critical natural capital, albeit it is not strictly defined what capital is critical

Question: Which of the following terms can be defined as “irreversible transformation of a resource by loss of its utility or functionality”?

- A Resource dissipation
- B Resource depletion
- C Resource consumption
- D Resource use

Question: Which of the coal types is the highest-ranking (lowest % content in volatile matter and highest in carbon content)?

- A Peat
- B Lignite
- C Bituminous coal
- D Anthracite

Group 15:

Question: According to Gibson, what is the goal of sustainability assessment?

- A) To focus solely on ecological impacts
- B) To integrate social, economic, and ecological considerations into decision-making
- C) To prioritize short-term economic gains
- D) To eliminate all environmental risks

Question: Which environmental issue is especially linked to natural gas production?

- A Methane leaks
- B Mercury emissions

- C Sulfur dioxide from combustion
- D Radioactive waste

Question: Which of the following is **NOT** a sustainability theory?

- A) Three pillars
- B) Double diamond**
- C) Nested sustainability
- D) Triple bottom line

Group 52:

Question: What is NOT an example of a biotic resource?

- A: Plants
- B: Animals
- C: Water**
- D: Microorganisms

Question: What is most correct about epistemic uncertainty?

- A: Randomness
- B: Instrument error
- C: Lack of knowledge**
- D: Natural variation

Question: What is the definition of a by-product?

- A: The material recovered in connection with the extraction/processing of the primary (main) material**
- B: A material that is deliberately created as the main goal of production
- C: A raw material that is used before the main production process begins
- D: A resource which cannot be found in nature, it only exists through human production.

Group 60:

Question: What is social economic metabolism?

- A: A model used in industrial ecology to optimize industrial systems by matching waste outputs with resource inputs across processes
- B: Research paradigm that explores the turnover and processing of resources such as materials, energy, and labour at a societal level**
- C: A framework for designing closed-loop manufacturing systems that reduce the need for raw material extraction
- D: An assessment tool that evaluates the environmental impact of individual industrial processes through material flow analysis

Question: In which sustainability theory is natural and manufactured capital and considered essentially substitutable?

- A) Strong Sustainability
- B) Nested Sustainability
- C) Weak Sustainability**
- D) Absolute Sustainability

3. Question: Which of the following groups contains resources that are both Abiotic AND Renewable?

- A) Animals, Metals
- B) Wind, Ground Water**
- C) Wind, Microorganisms
- D) Solar Radiation, Fossil fuels

Group 30:

Question: What is the main idea of Industrial Ecology?

- A) To mine more fossil fuels.
- B) To copy natural ecosystems for industry**
- C) To focus only on consumer behavior
- D) To replace all natural systems with factories

Question: What is the purpose of MFA?

- a) To measure money in economy
- b) To study the flows and stocks of materials in a system**
- c) To calculate energy prices
- d) To predict the weather

Question: Which of the following is an example of a bioresource?

- a) Copper ore
- b) Wheat straw**
- c) Natural gas
- d) sand

Group 27:

Question: In Material Flow Analysis (MFA), what does *circularity* primarily refer to?

- A: Creating new materials from scratch
- B: Exporting materials outside the system
- C: **Cycling materials back into existing processes**
- D: Disposing of materials in landfills

Question: In Material Flow Analysis (MFA), what does the term *stock* refer to?

- A The total amount of materials imported into a country
- B **The amount of a material stored in a process or reservoir**
- C The speed at which a material moves through a system
- D The transfer of a substance into a specific output good

Question: A waste management company conducts a Material Flow Analysis (MFA) of its operations. The system includes the following processes: waste collection, sorting, incineration, recycling, and landfill. The company decides to set a temporal boundary of one year and a spatial boundary defined by the municipal area.

Which of the following flows would NOT be included within the defined system boundary?

- A) Household waste collected within the municipality.
- B) **Recyclable materials exported abroad for further processing.**
- C) Residues sent to a local landfill inside the municipality.
- D) Waste incinerated at the municipal incineration plant.

Group 10:

The ecological footprint is....

- A: CO₂ pollution that is released into the atmosphere.
- B: how much resources are left for humans on earth.
- C: **a measure of how much land and resources humans use to produce compared to what the earth can regenerate.**
- D: the state of animal life on earth.

Question: Where does the metal in metallic ores come from?

- A: **The Earth's crust or mantle**
- B: Asteroids
- C: Meteors
- D: The Earth's core

Question: If a material has a circularity rate of 60%, what does that mean?

- A: 60% of the material is lost during processing
- B: **60% of the demand is met through recycled material**
- C: 60% of virgin resources are used
- D: 60% of waste is incinerated

Group 33: (August, Mohammad, Gustav, Rasmus)

Question: What is the definition of uncertainty in mass flow analysis?

A: Quantification of the doubt about the measurement result*

B: The difference between the measured value and the 'true value' of the object being measured

C: How uncertain is the creator of the analysis in presenting his/her findings?

Question: In mass flow analysis, drinking water would be defined as a:

A: Good*

B: Substance

C: Byproduct

Question: What defines a black box process in MFA?

A: A process that contains more than two subprocesses and only the input/output is of interest*

B: A process that isn't visible to humans, e.g. happens inside the steel tank.

C: A process that doesn't meet theoretical expectations

Group 13:

Question: What are the three pillars of the Three Pillars sustainability theory?

A) People, Planet, Profit.

B) People, Economy, Profit.

C) Environment, Economy, Society.

D) Society, Economy, Industry.

Question: What is the definition of a mineral?

A) Elemental substance, typically with metallic properties.

B) Naturally occurring inorganic solid with a definite and crystalline structure.

C) Naturally occurring solid composed of one or more minerals or mineraloids.

D) Rock or sediment containing sufficient concentration to be economically extracted.

Question: What is the definition of an ore?

A) Elemental substance, typically with metallic properties.

B) Naturally occurring inorganic solid with a definite and crystalline structure.

- C) Naturally occurring solid composed of one or more minerals or mineraloids.
- D) Rock or sediment containing sufficient concentration to be economically extracted.**

Group 40:

Question: Which sector is the main contributor to the EU material footprint?

- A Transportation sector
- B Housing (building sector)**
- C Energy sector
- D Agriculture sector

Question: In the concept of “nested sustainability,” how are the three spheres arranged?

- a) Society contains the economy, and the environment is inside the economy
- b) Economy contains society, and society contains the environment
- c) Economy is constrained by society, and society is constrained by the environment**
- d) Environment is contained by the economy and society equally

Question: What does “Earth Overshoot Day” represent?

- a) The day Earth travels farthest from the sun
- b) The day when humanity has used all the resources Earth can renew in one year**
- c) The day global recycling is celebrated
- d) The day fossil fuels run out

Group 42:

Question: **Which definition fits the term "Fund resources"?**

- A) Resources that exist as finite fixed amounts in the natural environment, With renewal rates beyond human timescales (e.g. coal, oil, mineral ores)
- B) Resources that exist as funds in the bio- or technosphere, but which current capabilities to access or utilize them (in a sustainable or economically viable manner) are insufficient (e.g. rare earths in deep sea floors, landfill mining).
- C) Resources that can be depleted at a rate depending on the ratio between extraction and renewal. Both permanent depletion and exhaustion of the resource may occur (e.g. plants, animals, groundwater)**
- D) Resources that can be used, but not depleted, although local or temporal non-availability may occur (e.g. freshwater precipitation, solar radiation, wind)

Question: **How is the Circularity Material Use Rate (CMU) generally calculated in the European Union?**

- A) CMU uses the mass of recycled solid waste from waste treatment facilities divided by the ratio of **Energy consumption per ton of** domestically consumed materials as an indicator of resource efficiency.
- B) CMU is calculated by dividing the **sum of all In-use stocks and** recirculated materials (U) by the sum of the Domestic Material Consumption (DMC) and (U)

- C) CMU is defined as the ratio of Recycled waste in treatment operations (U) to all domestically consumed materials (DMC), which includes all resource extractions and imports while excluding exports.
- D) CMU rate can be estimated by dividing the total domestic material consumption (DMC) (excluding exports) by the total mass of recycled materials (excluding thermal recycling).
- E) CMU rate is the ratio between all recycled materials in treatment operations (U) and the sum of U and the Domestic Material Consumption (DMC), which includes all resource extractions and imports while excluding exports.

Question: **Which process best describes the natural formation of coal as a fossil fuel?**

- A) The remains of marine microorganisms are buried under sediments, subjected to heat and pressure, eventually forming.
- B) Plant material accumulates in swampy environments, forming peat, which over millions of years transforms under heat and pressure into brown coal (lignite), black (bituminous) coal, and anthracite.
- C) Plant vegetation in high altitudes dies off during prolonged periods of repeated glacial freezing and melting, forming peat, which over millions of years transforms under heat and pressure into lignite, bituminous coal, and anthracite.
- D) Sediments of organic material are buried under layers of rock by geological processes and transformed into graphite under high pressure and heat from tectonic movement.
- E) In tectonic convergence zones sediments of sulfite debris and volcanic ash compact plant debris and peat under extremely hot, anaerobic conditions into lignite, bituminous coal and eventually anthracite

Group 16:

1. What resource is non-renewable?

- A. Freshwater
- B. Wind
- C. Metals (right answer)
- D. Plants

2. Which of these analytical methods is specifically concerned with quantifying the material and energy flows within cities, treating them as complex ecosystems?

- A. Environmentally Extended Input-Output Assessment
- B. Urban Metabolism (right answer)
- C. Industrial Symbiosis
- D. Socio-economic Metabolism

3. How did the technological and resource focus of the First Industrial Revolution differ from that of the Second Industrial Revolution?

A. The First was centered on mechanical production using steam power, while the Second advanced to using steel, electricity, and petroleum. (right answer)

B. The First was based on digital devices and AI, while the Second was focused on computers and plastics.

C. The First relied on chemicals and rubber, while the Second utilized wind and solar energy.

D. The First laid the intellectual groundwork for progress, while the Second was defined by basic manufacturing and textiles.

Group 25:

1. Which of the following is an example of a stock resource?

A. Freshwater precipitation

B. Wind

C. Coal (right answer)

D. Solar radiation

2. What is the main environmental concern associated with methane leaks from natural gas production?

A. Soil contamination

B. Groundwater salinization

C. Global warming and climate change (right answer)

D. Habitat loss.

3. Which type of uncertainty arises from a lack of knowledge, such as missing or ambiguous data?

A. Aleatory uncertainty

B. Epistemic uncertainty (right answer)

C. Parameter uncertainty

D. Scenario uncertainty

Group 57:

1. What is parameterization in MFA

a. Using raw data and formulas instead of computed numbers in unit process datasets.

b. Replacing system boundaries with arbitrary assumptions to simplify calculations.

c. Converting all data into percentages instead of absolute values.

d. Aggregating different material flows into one single flow for easier visualization.

2. What is the main difference between *error* and *uncertainty* in measurements?

a. Error refers to the randomness in data, while uncertainty refers to mistakes in calculation

b. Error is the difference between measured and true value, while uncertainty is the doubt about the measurement result

c. Error is only present in poor measurements, while uncertainty is eliminated in careful measurements

- d. Error refers to variability in scenarios, while uncertainty is related to system design
3. Which of the following lists correctly shows the three main types of rocks based on their origin?
- a. Igneous, Sedimentary, Metamorphic
 - b. Primary, Secondary, Tertiary
 - c. Granite, Gypsum, Marble
 - d. Volcanic, Fossil, Crystal
4. Which of the following best describes *epistemic uncertainty*?
- a. Randomness and variability in the data due to natural fluctuations
 - b. Uncertainty arising from measurement errors and limited sample size
 - c. Lack of knowledge, ambiguous data, or approximations in models
 - d. Uncertainty caused by choices and controversies in scenario comparisons

Group 11:

- 1. The "Three Pillars" of sustainability are most commonly defined as:
 - a. People, Planet, Profit
 - b. Environment, Economy, Society
 - c. Efficiency, Equity, Ecology
 - d. The environment contains society, which contains the economy.
- 2. According to the concept of "Weak Sustainability":
 - a. Natural capital and manufactured capital are largely substitutable.
 - b. Natural capital is unique and cannot be replaced by manufactured capital.
 - c. The primary goal is to maximize profit above all else.
 - d. Social equity is more important than environmental protection.
- 3. Which statement is most accurate regarding recent global ecological overshoot and the ecological footprint metric?
 - a. Earth Overshoot Day fluctuates around 1 Earth and is trending downward.
 - b. Earth Overshoot Day has plateaued near ~1.75 Earths, indicating persistent overuse despite some progress.
 - c. Ecological overshoot is not relevant for material policy.
 - d. The ecological footprint omits carbon uptake capacity.

Group 20:

- What is the difference between a bottom-up and top-down approach to life cycle engineering?
- Bottom-up is when you look at a specific life cycle process and improve it in isolation, whereas top-down approach considers overall planetary boundaries and limitations before coming up with solutions to stay within those limits.
 - Bottom-up is about looking at a specific problem and fixing it, and top-down is about extending the life of in-use materials.

- Bottom-up is about decreasing the material use by improving recycling, whereas top-down approach considers overall planetary boundaries and limitations before coming up with solutions to stay within those limits.
- Bottom-up is about decreasing the material use by improving recycling and top-down is about extending the life of in-use materials.

What is the difference between MFAs and SFAs?

- SFA focuses on a substance's chemical elements or molecules flow, whereas MFA considers several
- They are the same thing
- MFA is the main analysis while SFA is the subsequent, more detailed one
- MFA focuses on material flows while in production, SFA flows show life cycles after materials enter the economic market

What are the six biomass uses (6 Fs)?

- Feeding, Food, Fertilizer, Fuel, Feedstock and Fibre
- Feeding, Food, Fertilizer, Fuel, Fabric and Fibre
- Feeding, Food, Firing, Fuel, Fermentation and Fluorophore
- Feeding, Food, Fluorophore, Fuel, Feedstock and Fatty acids

Group 29

What does SDG stand for?

- A) Sustainable Development Goals
- B) Scientific Data Guidelines
- C) Strategic Development Group
- D) Social Development Grid

What's the difference between biotic and abiotic resources?

- A) Biotic resources are non-living, while abiotic resources are living
- B) Biotic resources are renewable, abiotic resources are always non-renewable
- C) Biotic resources come from living organisms, abiotic resources come from non-living sources
- D) Biotic resources are man-made, abiotic resources are natural

What does MFA stand for?

- A) Mineral Fossil Assessmentf
- B) Material Flow Analysis
- C) Mechanical Flow Architecture
- D) Mass Fraction Allocation

Group 50

Question 1: Uncertainty in an MFA that arises from natural variability or randomness is known as:

- A. Epistemic Uncertainty
- B. Scenario Uncertainty
- C. Aleatory Uncertainty
- D. Parameter Uncertainty

Question 2: Which of the following combination only includes goods?:

- A. drinking water, sewage, wood
- B. Carbon, gold, silver
- C. sewage, methane, solid waste
- D. fuel, oil, CO₂

Question 3: Which of the following is an example of an abiotic resource?

- A. Timber from a forest
- B. Natural gas
- C. Agricultural crops
- D. Fish stocks

Group 55

What are fossil resources primarily used for?

- A) Generating electricity from renewable sources
- B) Manufacturing food products
- C) Energy, transport, chemicals, and materials
- D) Growing organic matter
- E) Purification of water

Which statement about sankey diagrams is NOT true?

- A. Sankey diagrams show flows between nodes using arrows or bands whose width is proportional to the quantity.
- B. They are useful for visualizing how quantities move between different parts of a system.
- C. Sankey diagrams always show the exact sequence of events over time.
- D. They emphasize the magnitude of flows, making it easy to see major contributors.
- E. Sankey diagrams can display both inputs and outputs in a single overview.

In MFA which statement describes best the difference between a substance and a good?

- A) A substance is an economic entity of matter with value, while a good is a chemical element or compound.
- B) A substance is a chemical element or compound composed of uniform units, while a good is an economic entity of matter with positive or negative value.
- C) A substance and a good are the same, both referring to raw materials in MFA.

- D) A substance refers to waste materials only, while a good refers to useful products
- E) A substance is always natural, while a good is always man-made.

What is the main advantage of a Sankey diagram compared to a flow chart?

- A. It shows the logical sequence of steps in a process.
- B. It represents the amount or magnitude of each flow using arrow thickness.
- C. It highlights decision points and branching logic.
- D. It simplifies processes into symbols like rectangles and diamonds.
- E. It always uses only straight lines instead of curved or angled flows.

Group 18

What does the term “critical raw materials” refer to in the EU’s resource policy?

- A) Materials that only exist in fossil resources
- B) Materials with high economic importance and high supply risk
- C) Materials that can always be recycled without losses
- D) Materials that are only used in agricultural production

What is the primary difference between *stock* and *flow* in Material Flow Analysis (MFA)?

- A) Stock is input, flow is output
- B) Stock refers to the storage of materials, while flow refers to the movement of materials through a system
- C) Stock is always renewable, flow is always non-renewable
- D) Stock is measured only in economic units, flow only in physical units

What is the main purpose of *industrial symbiosis* within industrial ecology?

- A) To increase fossil fuel extraction efficiency
- B) To optimize profit by outsourcing production
- C) To exchange resources, energy, or by-products between industries to reduce waste
- D) To separate biological and technical resources into independent systems

Group 17

What is a stock in MFA?

- A. Transformation, transport, or storage of flows in an MFA
- B. Economic entities of matter with either a positive or negative value. Often consist of one or more of the most basic flow type in an MFA.
- C. Any (chemical) element or compound composed of uniform units
- D. Amount of materials stored in a process, this can be temporary or long term.
- E. A material needed in the MFA

What is scenario uncertainty?

- A. Ambiguity or controversy about how to quantify or compare scenario objectives
- B. Uncertainty about the relations and mechanisms being studied
- C. Difficulty within determination of how the model is created in different scenarios.
- D. Missing/wrong/ambiguous data
- E. Limited sample sizes causing scenario uncertainty

What is not an abiotic resource?

- A. Fossil fuels
- B. Water
- C. Living organisms
- D. Metals
- E. Nuclear fuels

Group 1

Which of the following is NOT one of the commonly recognized 8 principles of sustainability?

- A. Intergenerational equity — ensuring that development today does not compromise opportunities for future generations.
- B. Integration — balancing environmental, social, and economic considerations in decision-making.
- C. Precautionary principle — avoiding serious or irreversible environmental harm even when there is scientific uncertainty.
- D. Substitution — replacing natural capital with technological solutions whenever possible to maintain growth.
- E. Efficiency — using resources in ways that maximize benefits while minimizing waste and harm.

What is the CMUrate?

- A. The circular use of material over the total use of materials
- B. How much of a certain material is used
- C. The rate of which a recycled material is used up
- D. The carbon material use
- E. The rate calcium material depletion.

What is aleatory uncertainty?

- A. Uncertainty due to variability, randomness and data quality
- B. Systematic error, model faults, system definition.
- C. Measurement error, analytical imprecision, sample size

- D. Scenario uncertainty, uncertainty due to choice, ambiguity of how to quantify objectives
- E. Uncertainty due to user error, uncertainty about relations being studied.

Group 35

How many Earths of resources per year are needed to sustain current global consumption?

- A) 1.00
- B) 1.25
- C) 1.75**
- D) 2.50
- E) 3.00

Which list contains only abiotic resources?

- A) Minerals, Water, Solar radiation**
- B) Fossil fuels, Microorganisms, Plants
- C) Metals, Wind, Escherichia coli (E. coli)
- D) Fungi, Rocks, Nuclear fuels

What are the “6 Fs” of biomass usage?

- A) Feeding, Food, Fertilizer, Fuel, Feedstock, Fibre**
- B) Feeding, Food, Fertilizer, Fuel, Forestry, Fibre
- C) Feeding, Food, Fertilizer, Fuel, Feedstock, Fabrics
- D) Feeding, Food, Fertilizer, Fuel, Fossils, Fibre

Group 21

What does stock refer to in MFA terminology?

- A) A flow of materials moving per unit of time
- B) The amount of material stored in a process or reservoir**
- C) The boundary of a system in space and time
- D) The transfer coefficient of a substance into an output

Which sector uses the majority of global forestry biomass?

- A) Paper and pulp production
- B) Construction materials
- C) Fuel/energy use**
- D) Food production

What does Earth Overshoot Day represent?

- A) The day when global CO₂ emissions peak in a year
- B) The day when humanity's resource demand exceeds the Earth's yearly regenerative capacity
- C) The day when the UN reports global biodiversity loss
- D) The day when fossil fuel reserves are officially calculated